

## Week 4 Assignment

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### Objective:

Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory. Steps: Explore Azure Portal and create a Resource Group. Create a Storage Account, Blob Container, and upload a CSV file. Create Azure Data Factory and explore ADF UI (Author, Monitor, Manage). Create Linked Service (Blob Storage) and datasets (source and destination). Use Get Metadata activity to retrieve file information. Build a pipeline using Copy Data activity and configure source and destination. Execute the pipeline using Debug/Trigger and monitor execution. Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage. Implement an end-to-end pipeline (Blob → ADF → Destination) with metadata validation. Output: Screenshots (Resource Group, Storage, ADF, Pipeline) + pipeline execution results + brief summary.

### Resources:

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>  
[https://celebaltech.sharepoint.com/:w:/s/Celebal-LMS/IQDu6KrxRjt\\_Q4oeUYRJ1VcYATzbNIVoanpDrzZiW9VEIaY?e=2iujIU](https://celebaltech.sharepoint.com/:w:/s/Celebal-LMS/IQDu6KrxRjt_Q4oeUYRJ1VcYATzbNIVoanpDrzZiW9VEIaY?e=2iujIU)

## Below is the Step by Step Execution of all the Tasks

### Step 1: Create Resource Group

Created a new Resource Group named RG\_Assignment in Azure Portal. A Resource Group acts as a logical container that holds all Azure resources used in this project.

Home > Resource Manager | Resource groups

## Create a resource group

Basics Tags Review + create

**Resource group** - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription \* ⓘ Azure subscription 1

Resource group name \* ⓘ RG\_Assignment

Region \* ⓘ (US) East US

Previous Next Review + create

Figure: Step 1: Create Resource Group

## Step 2: Verify Resource Group

Verified that the Resource Group was successfully created and is available in Azure Resource Groups.

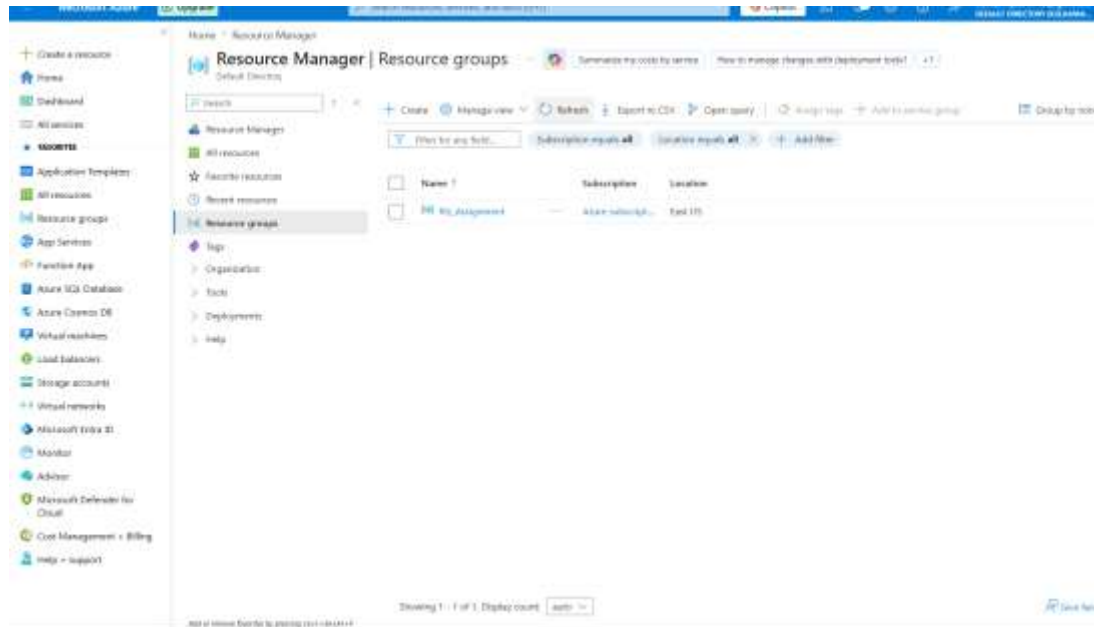


Figure: Step 2: Verify Resource Group

## Step 3: Create Storage Account

Created an Azure Storage Account that will be used to store the source and destination CSV files.

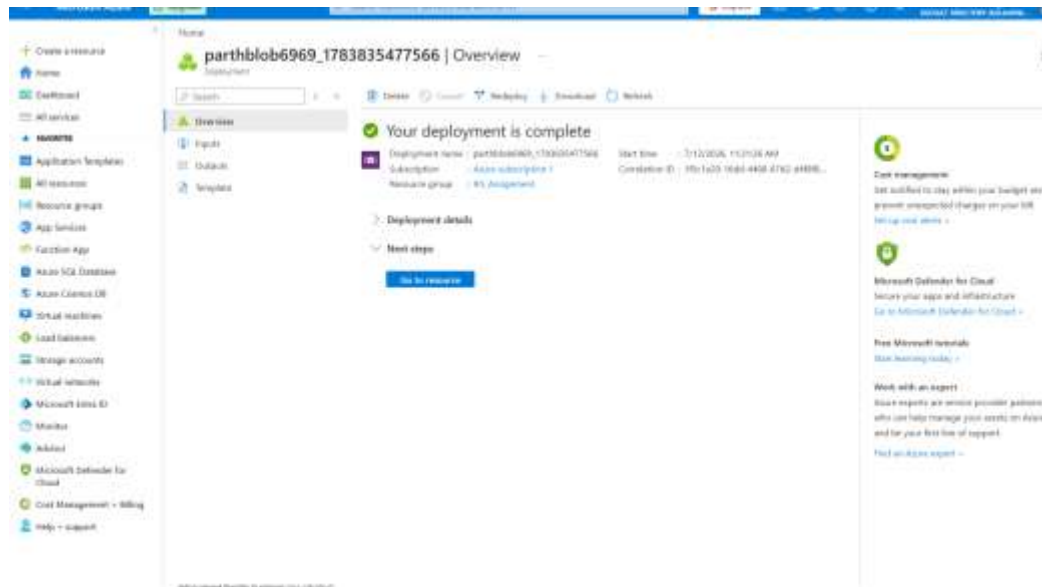


Figure: Step 3: Create Storage Account

## Step 4: Storage Account Deployment

Verified that the Storage Account deployment completed successfully.

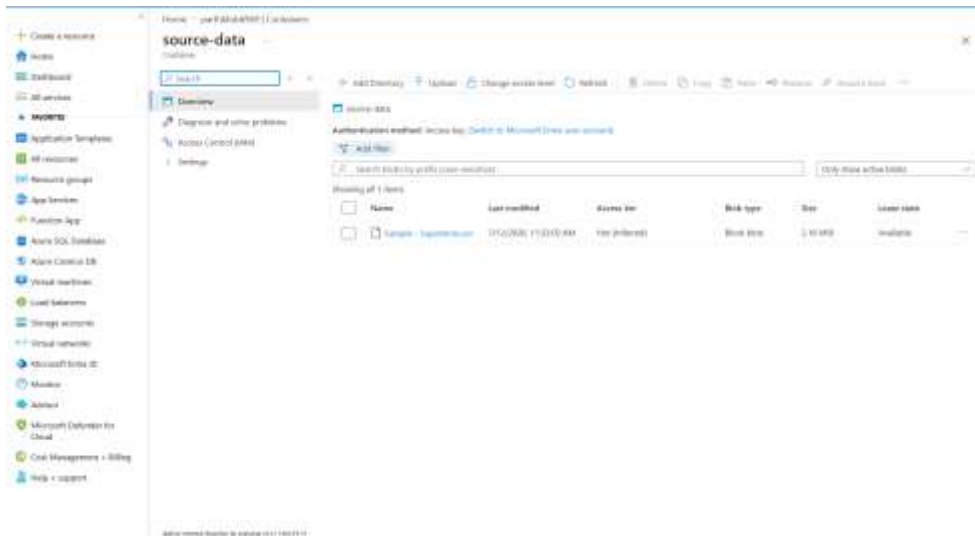


Figure: Step 4: Storage Account Deployment

## Step 5: Create Blob Containers

Created Blob containers named source-data and destination-data for storing input and output files.

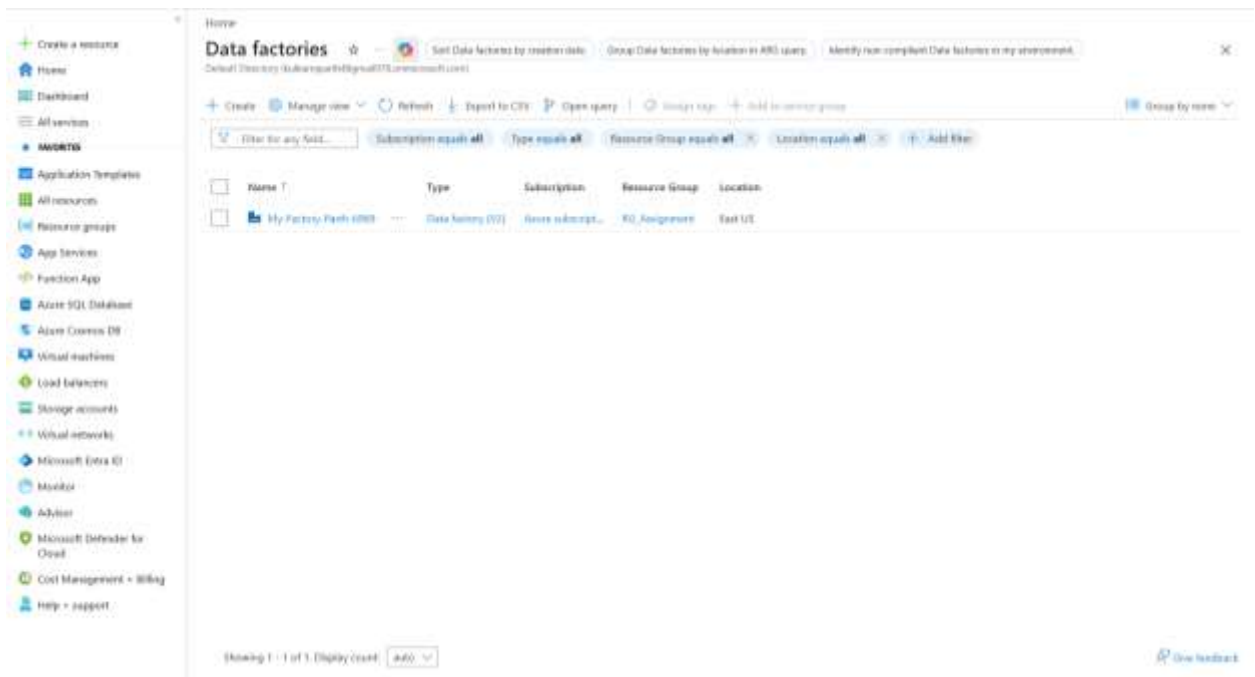


Figure: Step 5: Create Blob Containers

## Step 6: Upload Source CSV

Uploaded the Sample - Superstore.csv file into the source-data Blob container.

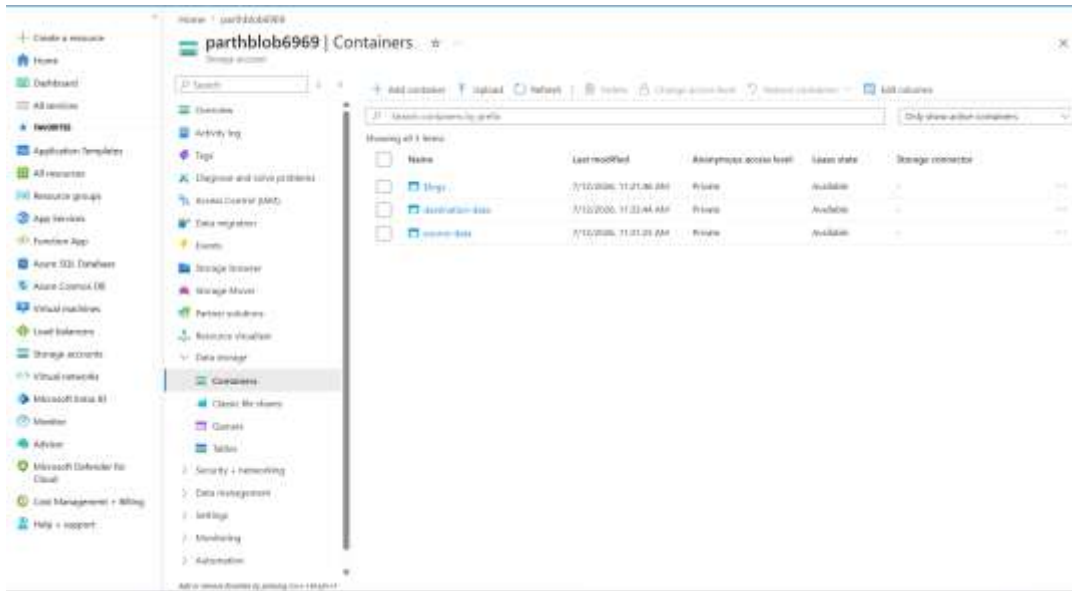


Figure: Step 6: Upload Source CSV

## Step 7: Create Azure Data Factory

Created an Azure Data Factory instance named My-Factory-Parth-6969.

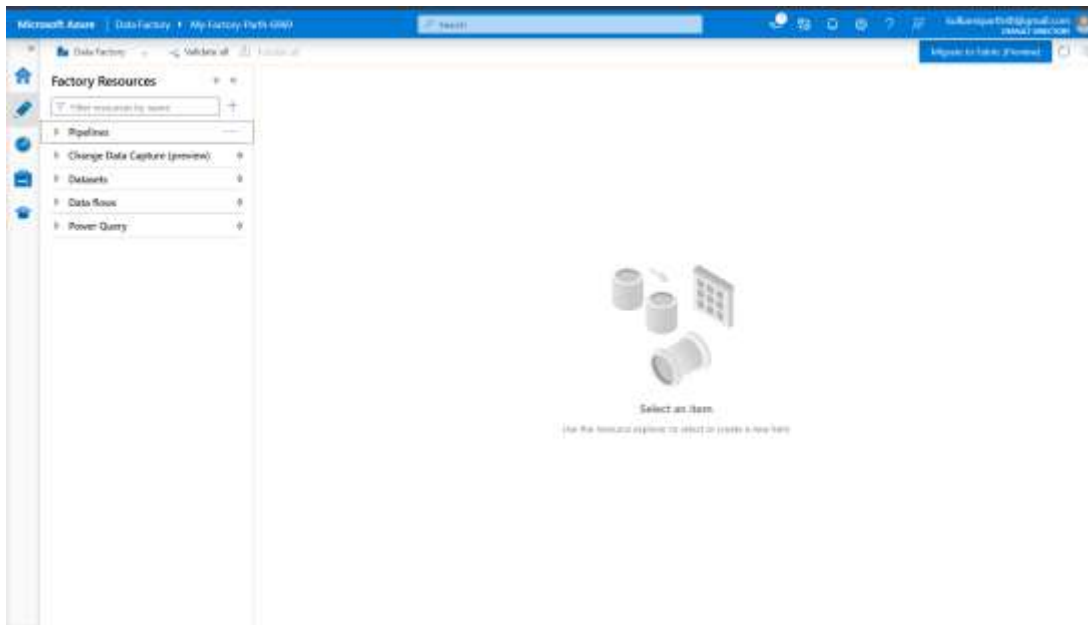


Figure: Step 7: Create Azure Data Factory

## Step 8: Open ADF Studio

Opened Azure Data Factory Studio to author and manage data pipelines.

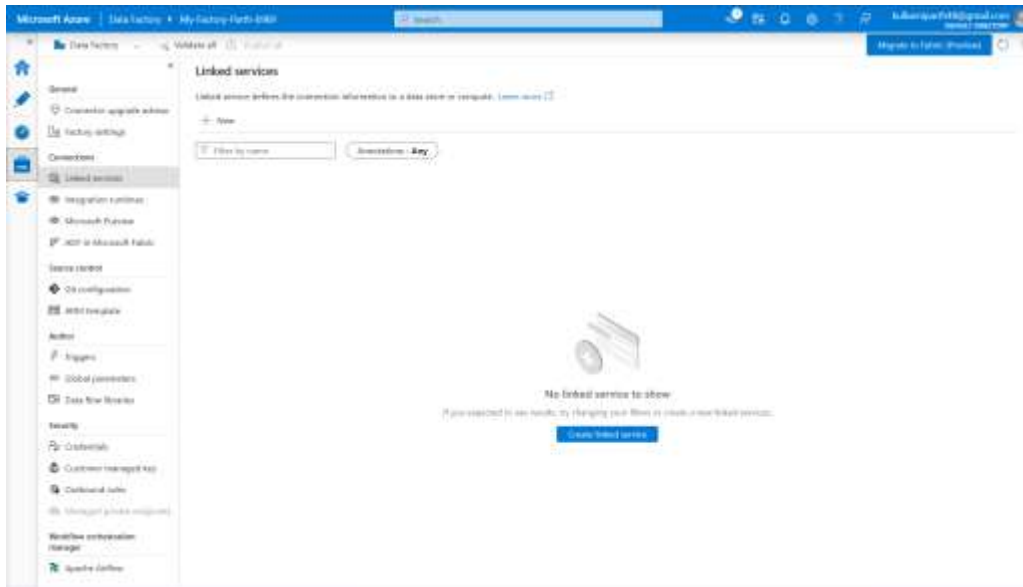


Figure: Step 8: Open ADF Studio

## Step 9: Create Linked Service

Created a Linked Service to connect Azure Data Factory with the Azure Blob Storage account.

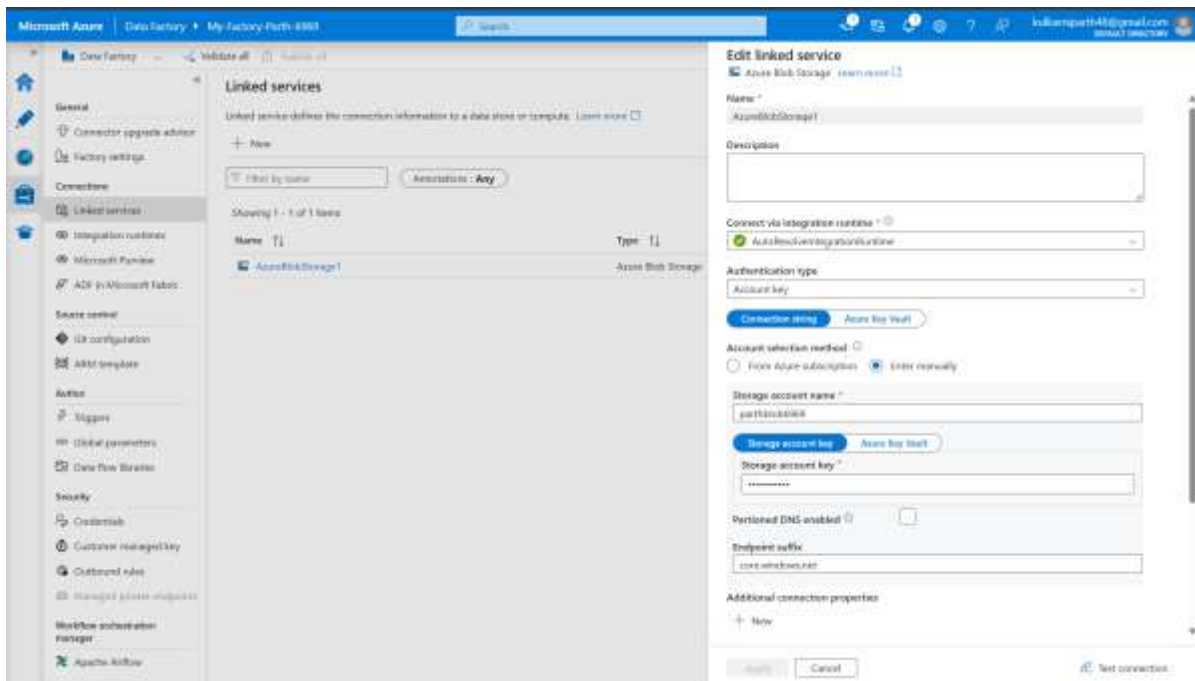


Figure: Step 9: Create Linked Service

## Step 10: Configure Linked Service

Configured the Azure Blob Storage linked service using Storage Account credentials and tested the connection.

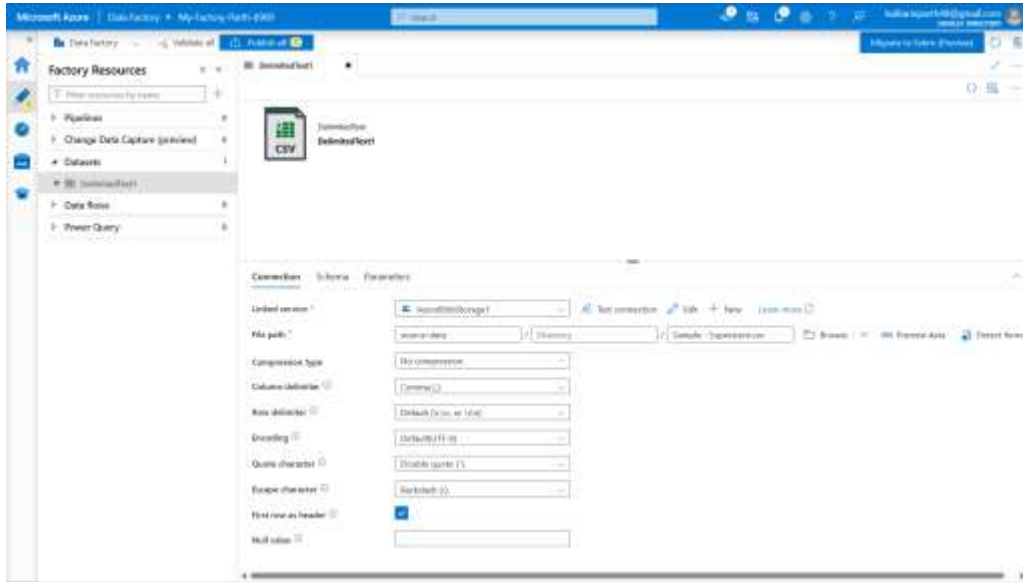


Figure: Step 10: Configure Linked Service

## Step 11: Create Source Dataset

Created a source dataset pointing to the Sample - Superstore.csv file in the source-data container.

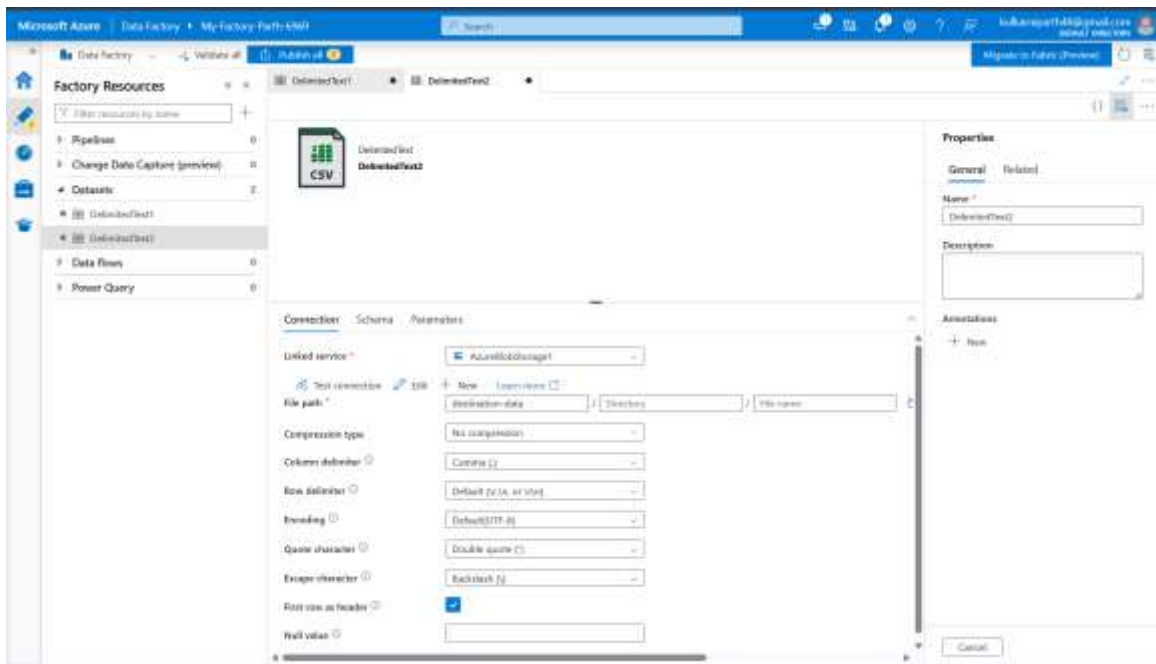


Figure: Step 11: Create Source Dataset

## Step 12: Create Destination Dataset

Created a destination dataset pointing to the destination-data container where the copied file will be stored.

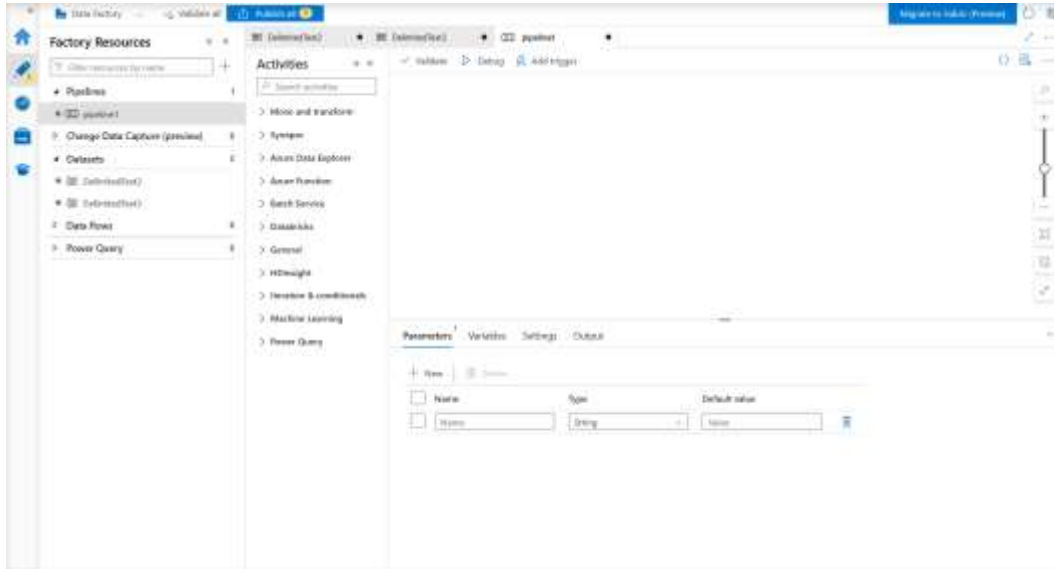


Figure: Step 12: Create Destination Dataset

## Step 13: Create Pipeline

Created a new pipeline in Azure Data Factory for implementing the data movement process.

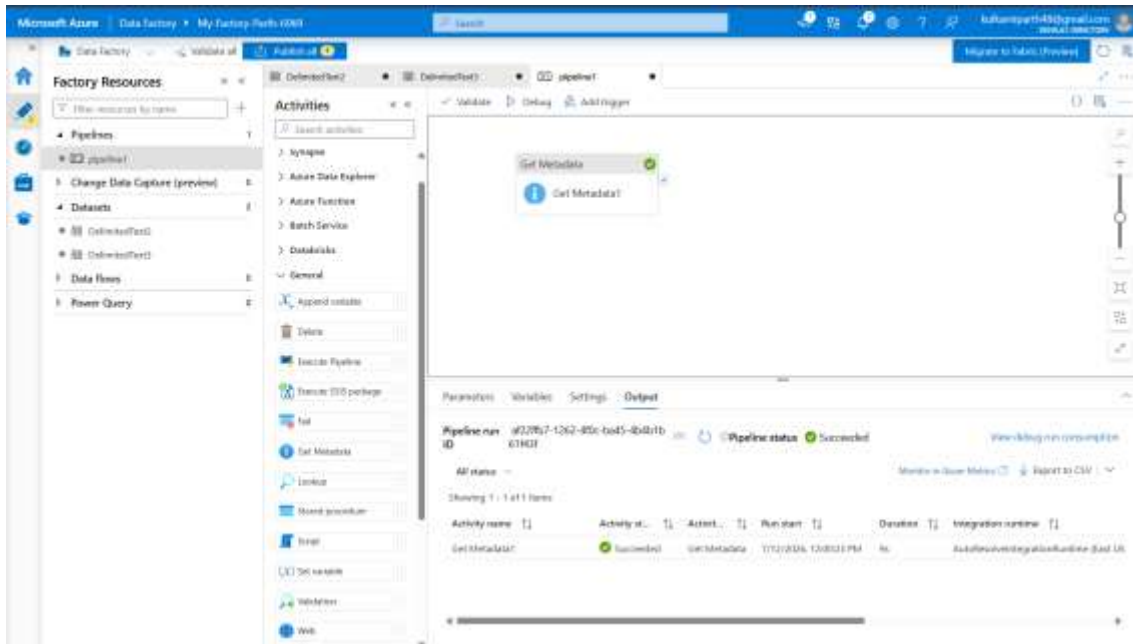


Figure: Step 13: Create Pipeline



## Step 14: Configure Get Metadata

Added the Get Metadata activity to validate the source file before copying.

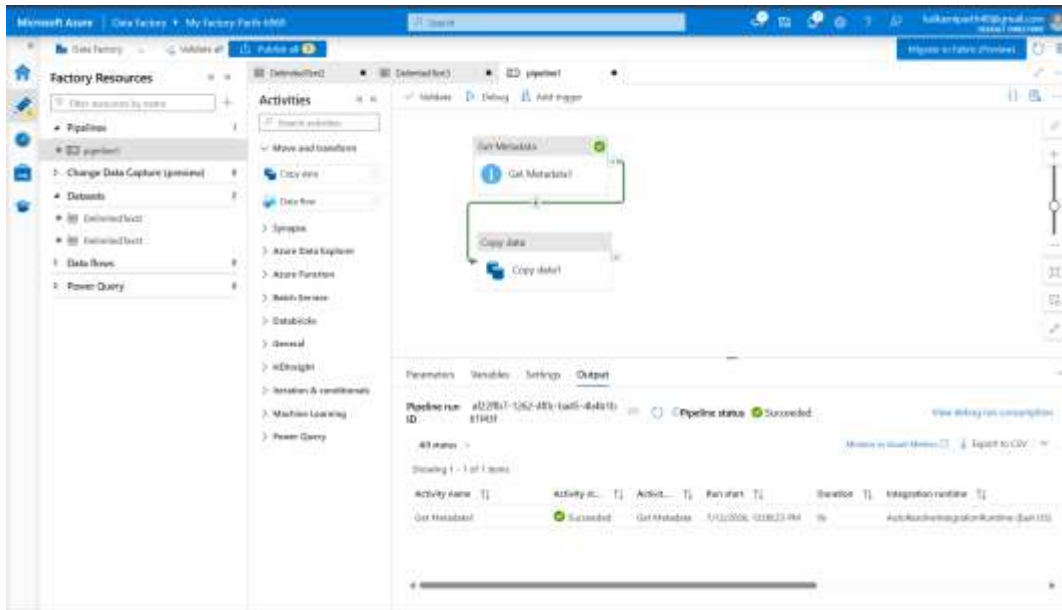


Figure: Step 14: Configure Get Metadata

## Step 15: Configure Copy Data

Added the Copy Data activity and connected it with the successful completion of Get Metadata.

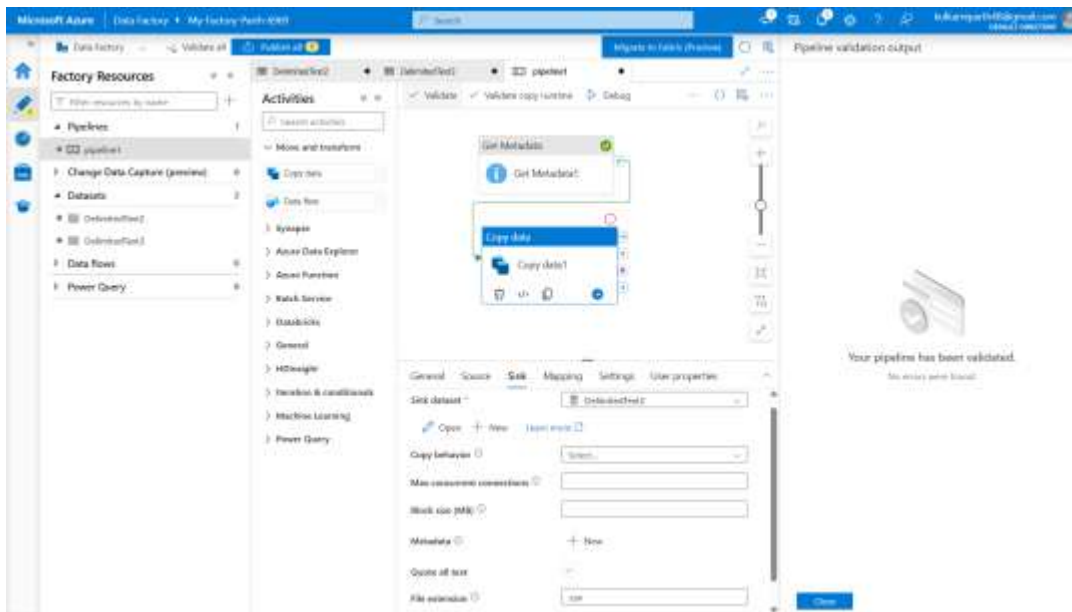


Figure: Step 15: Configure Copy Data

## Step 16: Validate and Publish

Validated the pipeline configuration and published all changes successfully.

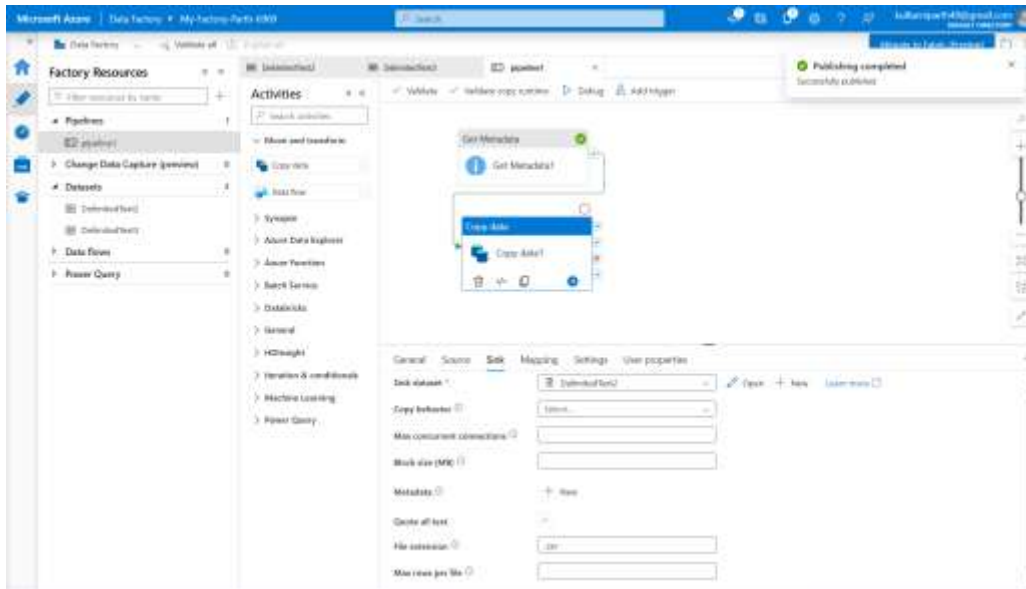


Figure: Step 16: Validate and Publish

## Step 17: Execute Pipeline

Executed the pipeline using Debug. Both Get Metadata and Copy Data activities completed successfully.

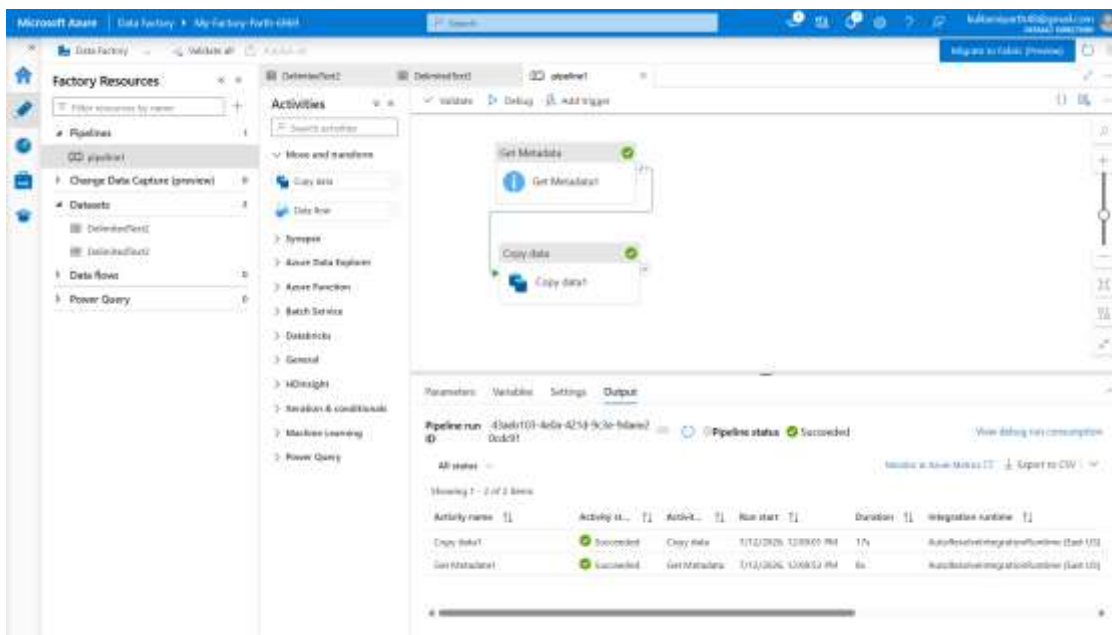


Figure: Step 17: Execute Pipeline

## Step 18: Verify Destination File

Verified that the CSV file was successfully copied into the destination-data Blob container.

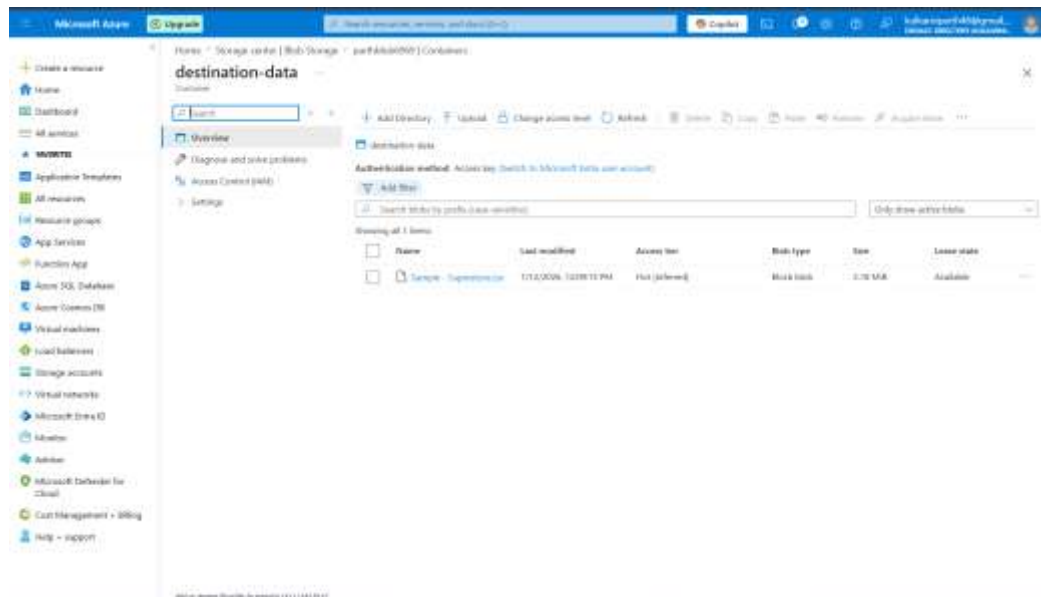


Figure: Step 18: Verify Destination File

## Step 19: IAM Role Assignment

Assigned the required Storage Blob Data Contributor role through Azure IAM to provide access for storage operations.

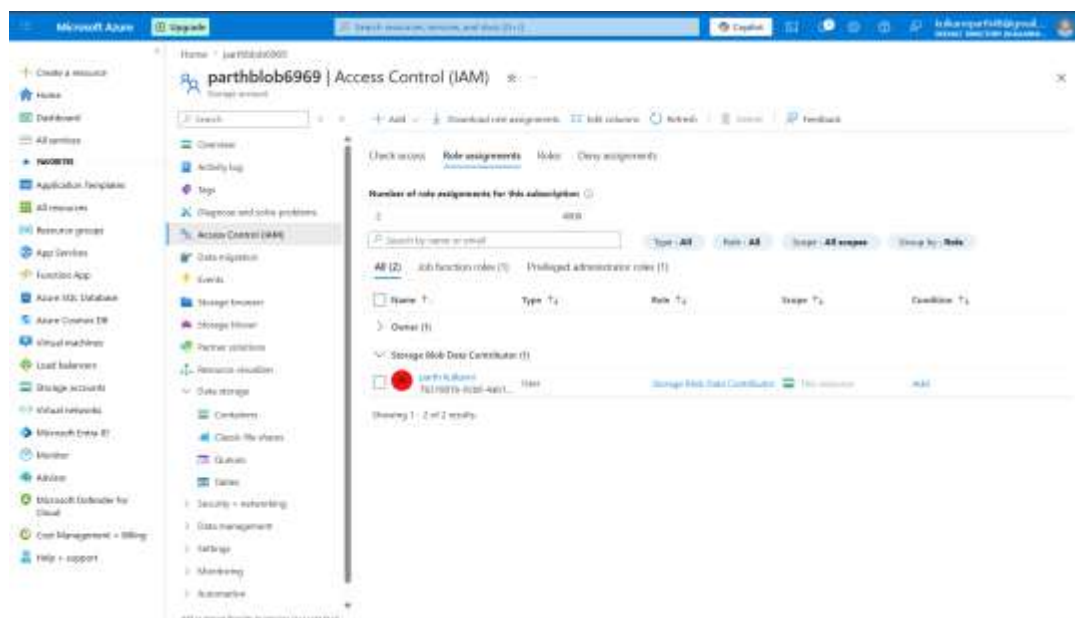


Figure: Step 19: IAM Role Assignment

## Conclusion:

This assignment provided practical experience in using Microsoft Azure cloud services to build an end-to-end data pipeline. A Resource Group and Azure Storage Account were successfully created, followed by the creation of Blob containers to store the source and destination CSV files. Azure Data Factory was configured by creating a Linked Service and datasets to establish connectivity with Azure Blob Storage.

The **Get Metadata** activity was used to validate the source file before execution, and the **Copy Data** activity successfully copied the CSV file from the source container to the destination container. The pipeline was validated, published, and executed successfully, and the execution was monitored through the Azure Data Factory Monitor. Finally, IAM role assignments were configured to ensure secure access between Azure Data Factory and Azure Storage.

Overall, this assignment helped in understanding the fundamental concepts of Microsoft Azure, Azure Blob Storage, Azure Data Factory, data movement, metadata validation, pipeline execution, monitoring, and access management. It provided valuable hands-on experience in building and managing cloud-based data integration solutions.